

Use-Case Flow Specification

Core Use Case: Submit and Automatically Evaluate Code Assignment
System: Automated Rubric Assignment Evaluator (PS #02)

Use Case ID & Name	UC-CORE-01: Submit and Automatically Evaluate Code Assignment
Primary & Secondary Actors	Primary: Student Secondary: Automated Sandbox Engine, Faculty Evaluator
Preconditions	1. Student is authenticated and enrolled in the course. 2. Assignment submission window is open. 3. Faculty has configured test suites and scoring rubrics.
Postconditions	Success: Student code is validated, test suite executes in sandbox, itemized rubric scores are logged and displayed. Failure: Submission is rejected with error diagnostics logged.

Main Success Scenario (MSS)

1. Student navigates to the assignment portal and uploads a compressed source archive (.zip/.tar.gz).
2. System receives the file, verifies archive structure, and extracts files into a secure staging directory.
3. System performs static syntax validation and dependency security checks (<<include>> UC-02).
4. System registers the submission into the automated evaluation queue with a unique tracking token.
5. Automated Sandbox Engine initializes an isolated container and mounts the sanitized student code.
6. Automated Sandbox Engine executes pre-configured unit test suites and performance benchmarks (<<include>> UC-04).
7. System captures test execution logs, stdout/stderr streams, and pass/fail metrics.
8. System calculates the itemized rubric score based on predefined faculty criteria weighting.
9. System commits results, logs, and timestamped rubric breakdown to the central database.
10. System notifies the student and renders the detailed rubric breakdown on the student dashboard.

Alternate and Exception Flows

3a. Invalid Archive Structure / Corrupted File

- 3a1. System detects corrupt zip header or unsupported compression format.
- 3a2. System rejects the upload and generates error alert: 'Invalid archive format. Please upload valid .zip/.tar.gz.'
- 3a3. System prompts student to re-upload. Flow returns to Step 1.

3b. Static Syntax or Import Rule Violation

- 3b1. Static analysis engine detects syntax errors or unauthorized external libraries.
- 3b2. System halts execution pipeline and marks status as 'Validation Failed'.

3b3. System generates syntax diagnostic log for the student. Flow terminates without grading.

6a. Execution Timeout / Infinite Loop in Sandbox

6a1. Code runtime exceeds the maximum sandbox timeout threshold (15 seconds).

6a2. Sandbox container forcibly terminates child execution processes.

6a3. System flags test cases as 'Timed Out' (0 points) and logs diagnostic trace.

6a4. System completes remaining evaluation criteria and resumes MSS at Step 8.